

# Energy Topology, Action, and the Homoclinic Boundary of the Duffing Flow

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29 August 2026 (revision 1)

## Abstract

We close an all-parameter phase-space atlas for the conservative Duffing oscillator  $\dot{x} = v$ ,  $\dot{v} = -\delta x - \beta x^3$  with  $\beta > 0$ . The energy levels are classified for both the single-well and double-well regimes, including the two homoclinic loops at the barrier. Turning points are roots of an explicit quadratic in  $x^2$ ; an endpoint-cancelled integral therefore gives the period and action on every regular oval, with the action derivative identity and center, quartic, and saddle limits. The ledger keeps the  $\beta = 0$  faces separate and treats the continuum of energy ovals as a source Hamiltonian family, not as isolated primitive cycles. Exact symbolic, independent numerical, replay, and hostile-mutation checks accompany the proof.

## 1 Model and invariant

Put

$$\dot{x} = v, \quad \dot{v} = -\delta x - \beta x^3, \quad H(x, v) = \frac{v^2}{2} + V(x), \quad V(x) = \frac{\delta x^2}{2} + \frac{\beta x^4}{4}, \quad (1)$$

where  $\beta > 0$  and  $\delta \in \mathbb{R}$ . The physical time is the common clock. Hamiltonian conservation follows immediately from  $H_x \dot{x} + H_v \dot{v} = 0$ . For a regular connected oval with turning points  $\ell < r$ , define

$$T(E) = \sqrt{2} \int_{\ell}^r \frac{dx}{\sqrt{E - V(x)}}, \quad I(E) = \frac{1}{\pi} \int_{\ell}^r \sqrt{2(E - V(x))} dx. \quad (2)$$

The endpoint square roots are integrable; the executable ledger evaluates (2) after  $x = (\ell + r)/2 + (r - \ell) \sin \theta/2$ .

**Theorem 1** (complete Duffing energy atlas). *Let  $\beta > 0$ . If  $\delta \geq 0$ , every  $E > 0$  is one smooth compact oval around the origin. If  $\delta < 0$ , write  $a = \sqrt{-\delta/\beta}$  and  $V_{\min} = -\delta^2/(4\beta)$ . Then  $V_{\min} < E < 0$  gives two ovals,  $E = 0$  gives two homoclinic loops to  $(0, 0)$ , and  $E > 0$  gives one outer oval. The squared turning points are*

$$y_{\pm} = \frac{-\delta \pm \sqrt{\delta^2 + 4\beta E}}{\beta}, \quad y = x^2. \quad (3)$$

*On each regular component, (2) is finite,  $I'(E) = T(E)/(2\pi)$ , and the period is positive. For  $\delta = -\alpha^2 < 0$ , the separatrices are*

$$x_h(t) = \pm \frac{\sqrt{2}\alpha}{\sqrt{\beta}} \operatorname{sech}(\alpha t), \quad v_h = \dot{x}_h. \quad (4)$$

*Moreover  $T(E) \rightarrow 2\pi/\sqrt{\delta}$  as  $E \downarrow 0$  when  $\delta > 0$ ,  $T(E) E^{1/4} \beta^{1/4}$  is constant when  $\delta = 0$ , and  $T(E)$  diverges logarithmically as a regular energy approaches the saddle level.*

*Proof.* The quartic potential is coercive, so regular compact components are the connected components described by the signs of the two roots in (3). For  $\delta < 0$  the identity

$$V(x) + \frac{\delta^2}{4\beta} = \frac{\beta}{4} \left( x^2 + \frac{\delta}{\beta} \right)^2$$

locates the two minima and the barrier. Separation of variables gives (2); the sine substitution removes endpoint singularities. Differentiating the action on a compact subinterval of regular energies and using the vanishing endpoint integrand gives  $I' = T/(2\pi)$ . At  $\delta = 0$ , the scaling  $x = E^{1/4}\beta^{-1/4}u$  proves the quartic law. The quadratic Taylor expansion at a nondegenerate center gives the center limit. Near the hyperbolic saddle,  $E - V(x)$  has a nondegenerate quadratic zero, and comparison with  $\int dx/\sqrt{\alpha^2 x^2 + |E|}$  gives the logarithmic divergence. Finally direct substitution of (4) into  $\ddot{x} - \alpha^2 x + \beta x^3 = 0$  proves the separatrix formula.  $\square$

## 2 Linear faces and source boundaries

The origin has linear rates  $\pm i\sqrt{\delta}$  for  $\delta > 0$  and  $\pm\sqrt{-\delta}$  for  $\delta < 0$ ; the well equilibria  $x = \pm a$  have rates  $\pm i\sqrt{-2\delta}$ . When  $\beta = 0$ , the positive- $\delta$  face is the harmonic oscillator, the negative- $\delta$  face is inverted and has no compact positive-energy ovals, and  $(\beta, \delta) = (0, 0)$  is free motion. These lower-dimensional models are not silently folded into the quartic theorem. The two inner components and the outer component are separated in the ledger, so a period near the barrier cannot be mistaken for a center period.

## 3 Action coordinates and evidence

On each regular component the action is strictly increasing because  $I'(E) > 0$ ; it is therefore a valid local action coordinate, while the homoclinic level is its logarithmic endpoint. The certificate contains five parameter cases and twenty energy rows, including both double-well components and the pure-quartic scaling control. A producer-independent checker makes 790 assertions, and SymPy verifies ten identities (conservation, factorization, turning polynomial, the profile residual, linear rates, and the action integrand). Clean replay is byte-for-byte and the hostile suite rejects twenty repaired-hash/schema mutations plus one stale-hash mutation (21/21 hostile checks).

## Source note and declarations

The phase-plane terminology follows Guckenheimer and Holmes [1]; period-function conventions follow Chicone [2]. The name Duffing is historical [3]; no priority claim is made. No prime or zero table, local arithmetic datum, Euler factor, root number, automorphy object, target functional equation, Hilbert–Pólya operator or Route-B input is used. Generative tools assisted drafting and code generation; the displayed claims are tied to the artifact chain. This is not external peer review. Scope literal: NO\_BAD\_EULER\_OR\_ROOT\_NUMBER.

## References

- [1] J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer, 1983. DOI: 10.1007/978-1-4612-1140-2.
- [2] C. Chicone, *Ordinary Differential Equations with Applications*, 2nd ed., Springer, 2006. DOI: 10.1007/0-387-35794-7.
- [3] G. Duffing, *Erzwungene Schwingungen bei veränderlicher Eigenfrequenz*, Vieweg, Braunschweig, 1918.